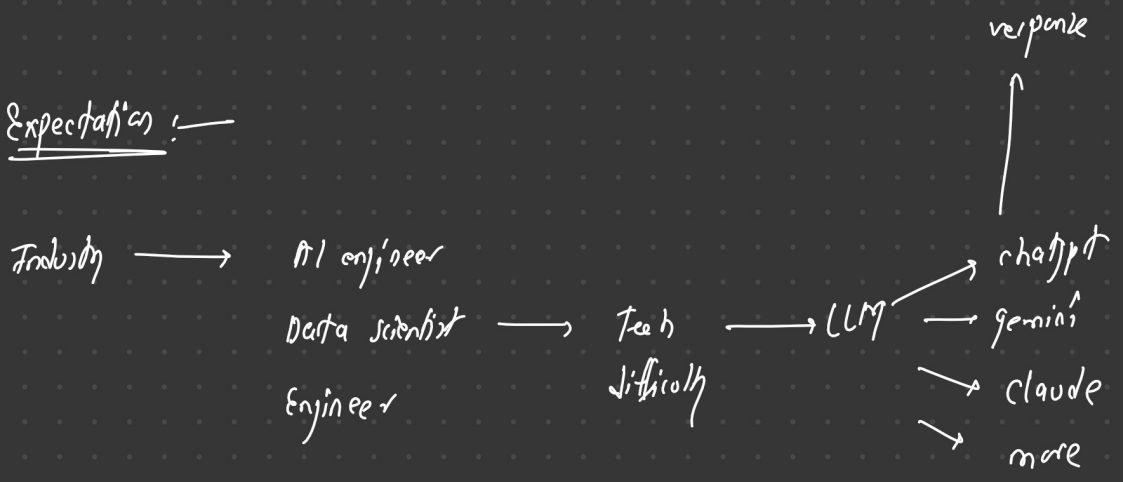
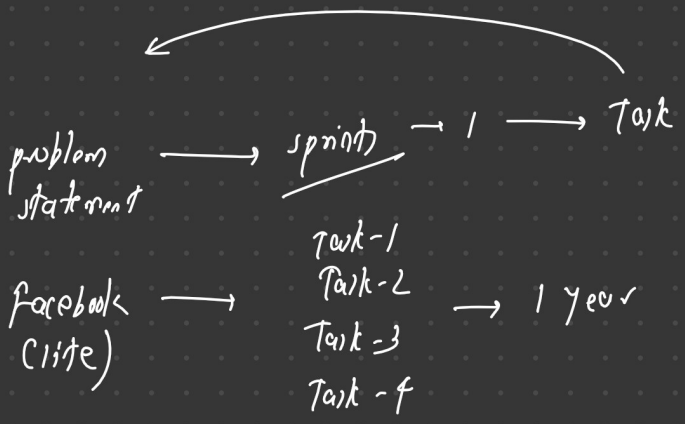
10-10-2026

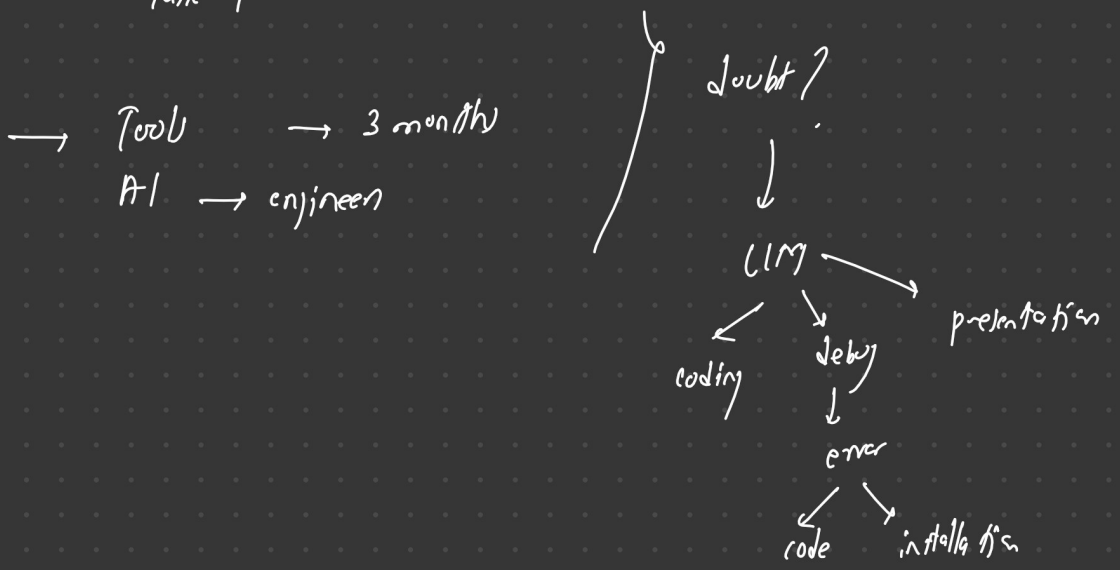
Agenda:

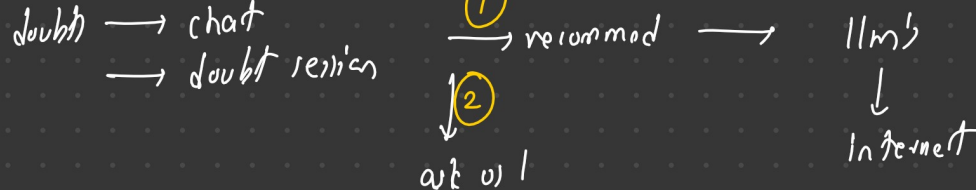
Tools + installation

Expectation :—









Task → open
→ port
→ server
→ display
→ zoom → software → code → RAM
→ CPU
→ chrome
→ interact ui

operating system (OS)
↓ interacting
UI

Command Line Interface (CLI)
↓
terminal → Tools

computer → screen → OS
→ CPU
↑
motherboard

Servers → powerful → process ⟷ Terminal
computer ↓
commands
↓
→ no UI
→ OS → linux

AI → python → programming language → code

code
↓
linux ⟷ Editor → write anything?
↓
config
↓
VS code

.py
.ipynb

Hybrid - IDE → editor
→ CLI
→ debug
→ more (apps)

vscode → run
→ edit
→ more

Pycharm → full IDE
↓
Heavy → Ram
→ storage
→ CPU

Python
① → python.org → single python version → 3.6
→ 3.14
② → Heavy
Anaconda / Miniconda → software → Tools → Conda
↳ python → 3.14
→ 3.6
→ 3.8
project manager
③ → lightweight
uv → software → Tools → uv
↳ python → 3.14
→ 3.6
→ 3.8
project manager

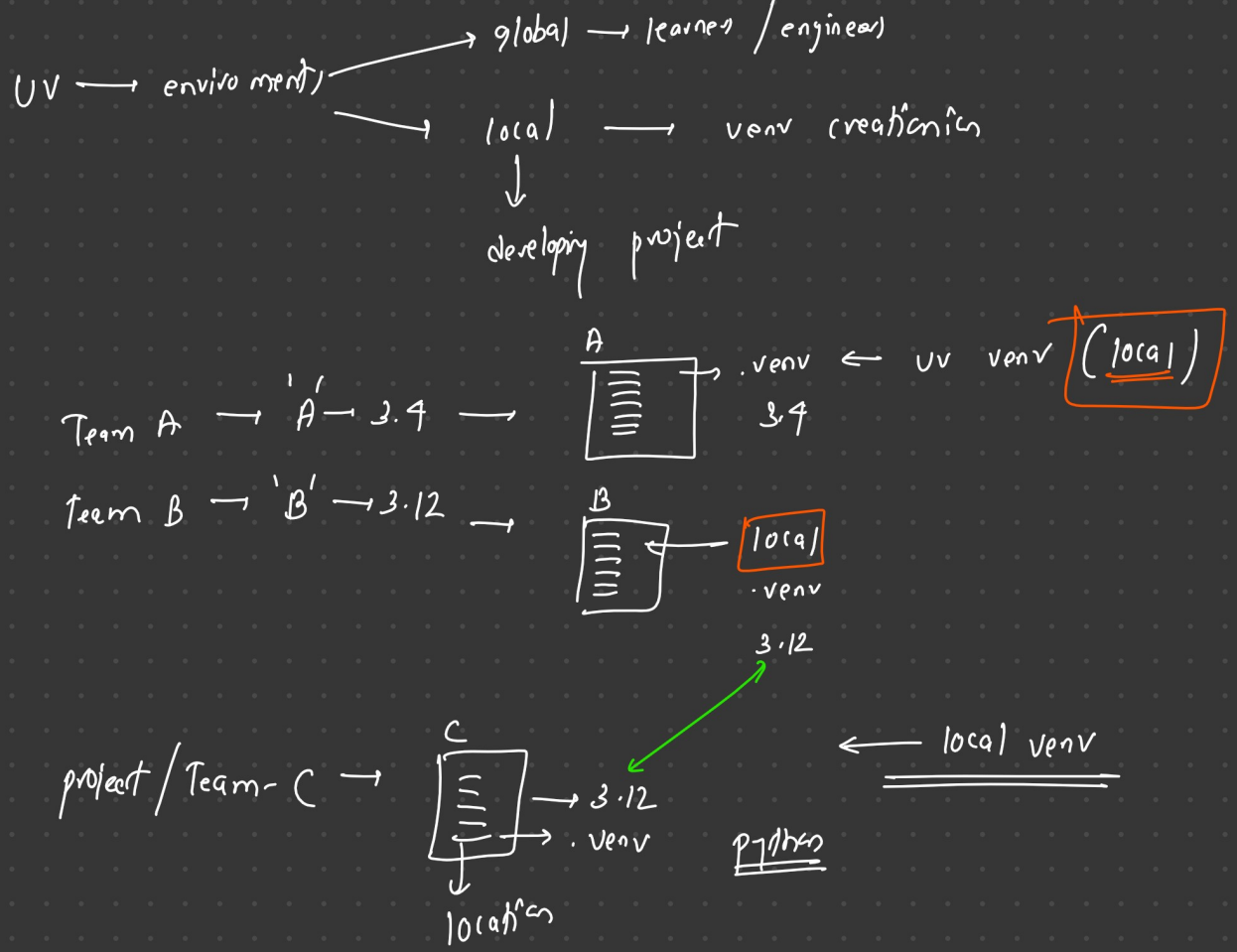
Team - A
(2014) → python - 3.4

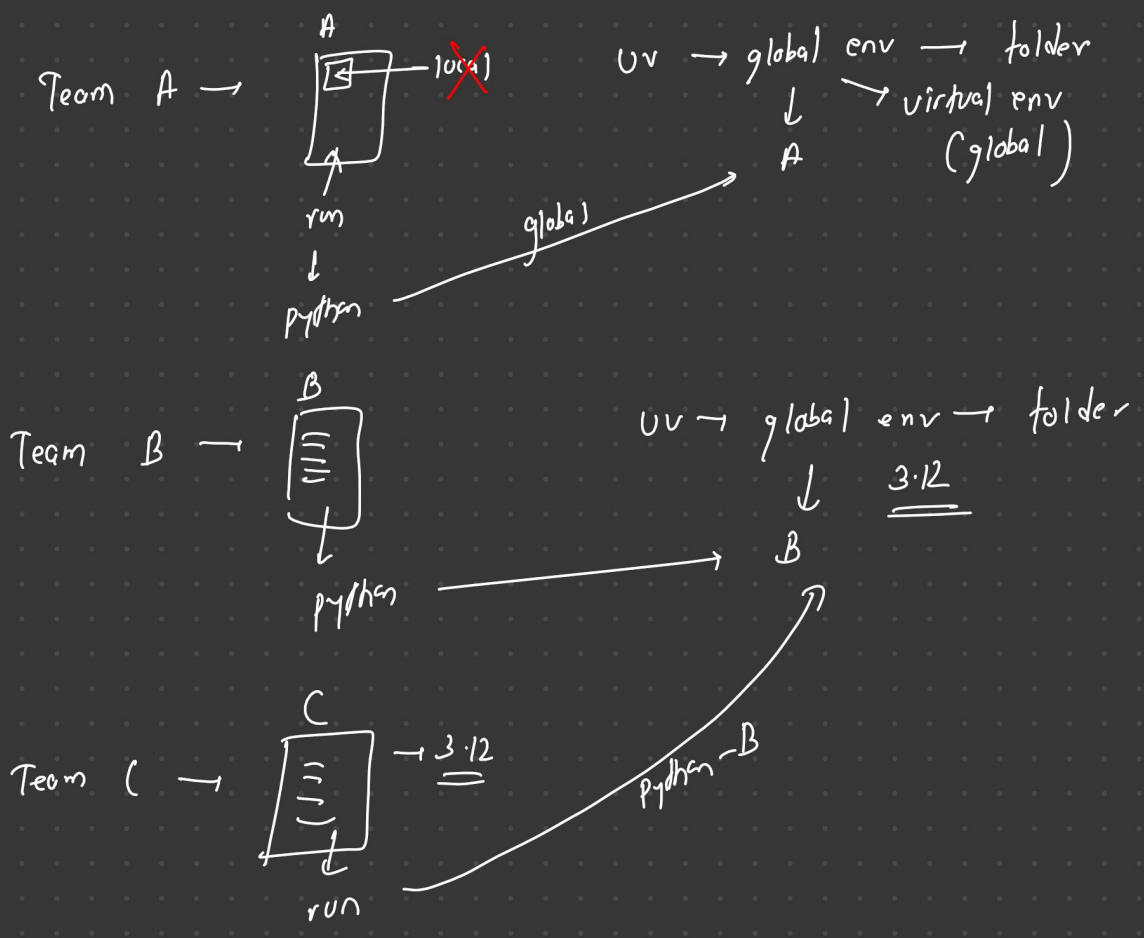
Team - B
(2026) (Agentic - AI) → Python - 3.12



name python version
(uv) → virtual env → A → 3.4 ← A
python
→ B → 3.12 ← B

Tools → CLI
→ vscode
→ uv
↓
pip PYPI


UV → environments
global → learners / engineers
local → venv creation in
developing project
Team A → 'A' → 3.4 →
A
.venv
3.4
UV venv (local)
Team B → 'B' → 3.12 →
B
local
.venv
3.12
project / Team-C →
C
3.12
.venv
location
local venv
Python


Team A →
A
local
run
python
global
UV → global env → folder
virtual env
A
(global)
Team B →
B
python
UV → global env → folder
3.12
B
Team C →
C
3.12
run
Python - B

Developer

A → local → .venv → 3.9

B → local → .venv → 3.12

C → local → .venv → 3.12

machine

global

A → 3.9

B → 3.12

A → run

B → run

C

machine

uv →